AI 时代的云原生与基础设施新生

Cloud Native in the Age of AI

张晋涛/Jintao Zhang

About

- 张晋涛 @ Kong Inc
- CNCF Ambassador
- Microsoft MVP（Most Valuable Professional）
- LFAPAC 布道师
- Kubernetes Ingress-NGINX maintainer
- KCD Beijing + vLLM 2026 组织者
- GitHub：https://github.com/tao12345666333
- zhangjintao@apache.org

AI 会不会让 IT 变得不再重要？

我们正处在一个“被 AI 重新定义”的时代

FOMO everywhere

这个问题，我们其实见过不止一次

Cloud Native is eating the world

Kubernetes 并没有消灭基础设施
它重塑了它

技术不会消失，它会被重新抽象

AI 时代 = 又一次「抽象跃迁」

2000s · 虚拟化

物理机

虚拟机

2014s · 云原生

虚拟机

Kubernetes

2024s · AI 时代

Kubernetes

AI Native Infra

我常被人问：云原生还值不值得搞？

Kubernetes
https://kubernetes.io › case-studies › openai

OpenAI Case Study

OpenAI began running Kubernetes on top of AWS
OpenAI runs key experiments in fields including robot

OpenAI
https://openai.com › index › scaling-kubernetes-to-750

Scaling Kubernetes to 7500 nodes

Jan 25, 2021 — We've **scaled Kubernetes clusters**
for large models like GPT-3(opens in a new window),

OpenAI
https://openai.com › index › scaling-kubernetes-to-250

Scaling Kubernetes to 2500 nodes

Jan 18, 2018 — We now operate several **Kubernete**
hardware), the largest of which we've pushed to over

consistently. Moreover, we discovered that enforcement methodology can change what the benchmark ends up actually measuring.

How we got here

We run Terminal-Bench 2.0 on a Google Kubernetes Engine cluster. While calibrating the setup, we noticed our scores didn't match the benchmark's official leaderboard, and infra error rates were surprisingly high: as many as 6% of tasks were failing because of pod errors, most of which were unrelated to the model's ability to solve the tasks.

The discrepancy in scores came down to enforcement. Our Kubernetes implementation treated the per-task resource specs as both a floor and a hard ceiling: each container was guaranteed the specified resources but killed the moment it exceeded them. Container runtimes enforce resources via two separate parameters: a guaranteed allocation—the resources reserved up front—and a hard limit at which the container is killed. When these are set to the same

不变的，是对“复杂性的管理”

AI 系统 = 更复杂的分布式系统

AI 系统挑战		分布式系统本质
模型路由 多模型选择 / A–B 测试 / 降级	≈	**Service Mesh** 流量治理 / 灰度发布 / 熔断
多 Agent 协调 任务拆分 / 状态同步 / 结果聚合	≈	**Orchestration** 工作流编排 / 状态机 / Saga
GPU 调度 显存管理 / 弹性伸缩 / 混合部署	≈	**Resource Scheduling** 资源配额 / 亲和调度 / 抢占
推理流量管理 请求排队 / 限流 / 实例间负载分发	≈	**Load Balancing** 负载均衡 / 限流 / 队列管理

KCD GROWING CLOUD NATIVE TOGETHER BEIJING vLLM

我们需要“AI Native 基础设施”

AI 不会取代 IT，它会让 IT 再次伟大

如果你想在 AI 时代保持竞争力：

理解系统，
分布式原理依然重要

掌握抽象，
API、Control Plane、Data Plane 不可或缺

拥抱变化，
AI 只是加速，不是颠覆

下一个十年，
属于那些既懂系统，又懂 AI 的人

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Thanks.